

TASK 2 – DEEP LEARNING IMAGE CLASSIFICATION PROJECT

Submitted By:

Katakam Priyanka

Role:

Artificial Intelligence Intern

Internship:

InternSpark AI Internship Program

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1. INTRODUCTION

Deep Learning is a subset of Artificial Intelligence that uses neural networks to learn complex patterns from data. Image classification is one of the most common applications of deep learning where the model predicts the category of an image.

In this project, a Convolutional Neural Network (CNN) was developed to classify images using the CIFAR-10 dataset.

2. OBJECTIVE

The main objectives of this project are:

- To understand deep learning concepts
 - To build a CNN model
 - To classify images into categories
 - To evaluate model performance
-

3. DATASET DESCRIPTION

Dataset Name:

CIFAR-10 Dataset

The dataset contains images belonging to ten categories:

- Airplane
- Automobile
- Bird
- Cat
- Deer
- Dog
- Frog

- Horse
- Ship
- Truck

Training Images: 50,000

Testing Images: 10,000

Image Size:

32 × 32 pixels

```
[1] import tensorflow as tf
from tensorflow.keras import layers, models

import matplotlib.pyplot as plt
import numpy as np

[2] (x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_data()

Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 1173s 7us/step

[3] print(x_train.shape)
print(x_test.shape)

... (50000, 32, 32, 3)
(10000, 32, 32, 3)

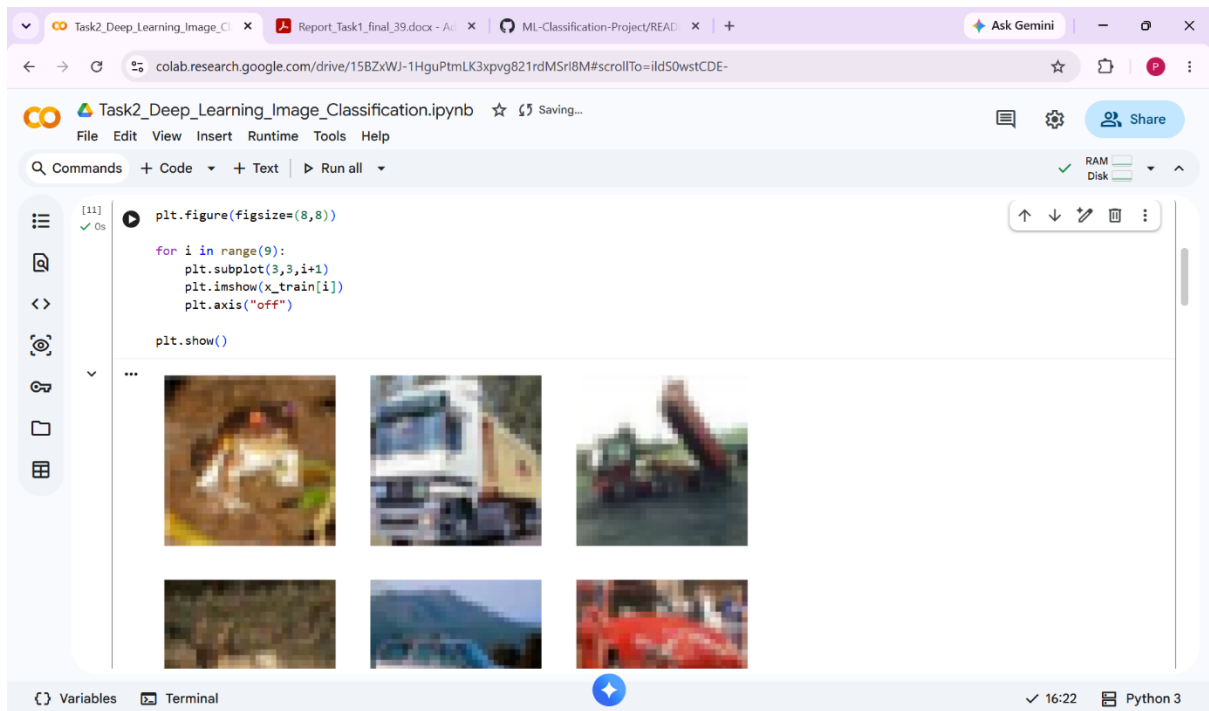
[11] plt.figure(figsize=(8,8))

for i in range(9):
    plt.subplot(3,3,i+1)
    plt.imshow(x_train[i])
    plt.axis("off")
```

4. DATA PREPROCESSING

Data preprocessing techniques performed:

- Loaded dataset
- Normalized image values
- Converted image pixels between 0–1
- Split training and testing data



5. MODEL ARCHITECTURE

The CNN model consists of:

Input Layer



Convolution Layer



Max Pooling Layer



Convolution Layer



Max Pooling Layer



Flatten Layer



Dense Layer



Output Layer

6. MODEL TRAINING

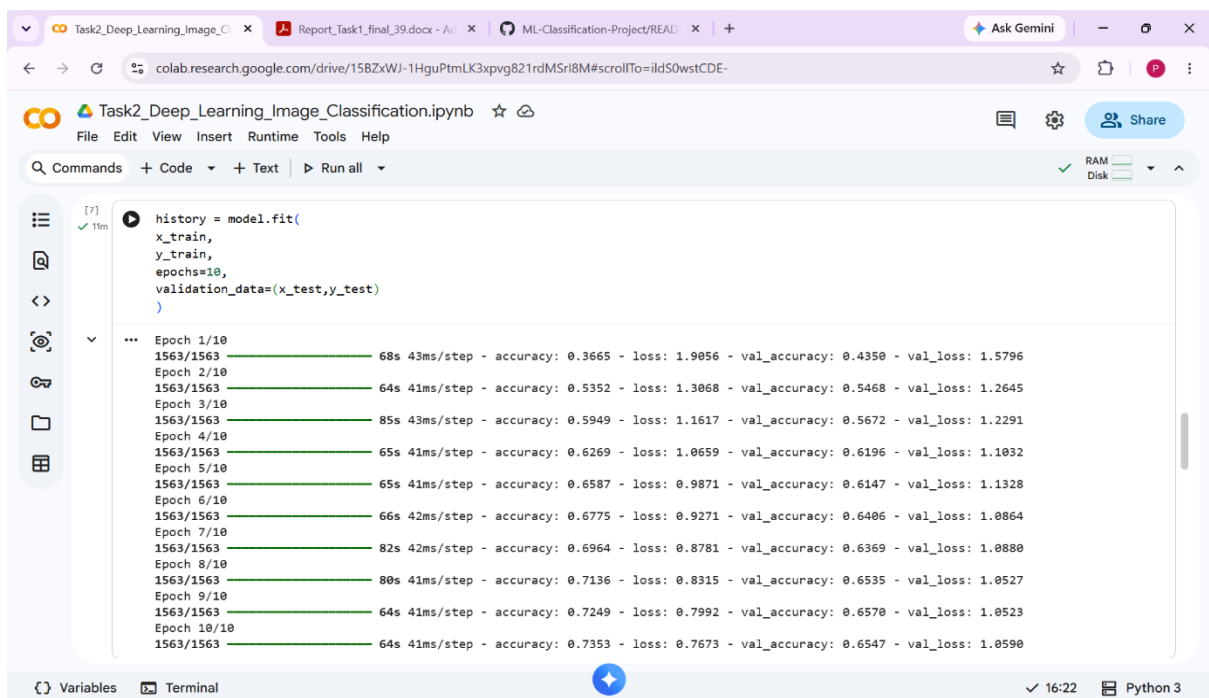
Model configuration:

Optimizer: Adam

Loss Function: Sparse Categorical Crossentropy

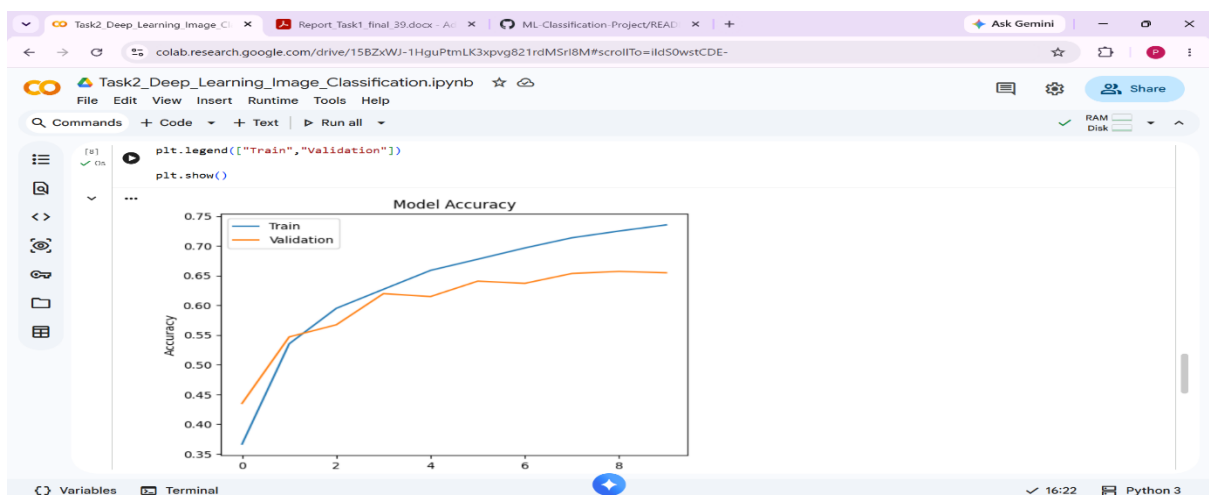
Epochs:10

Evaluation Metric: Accuracy



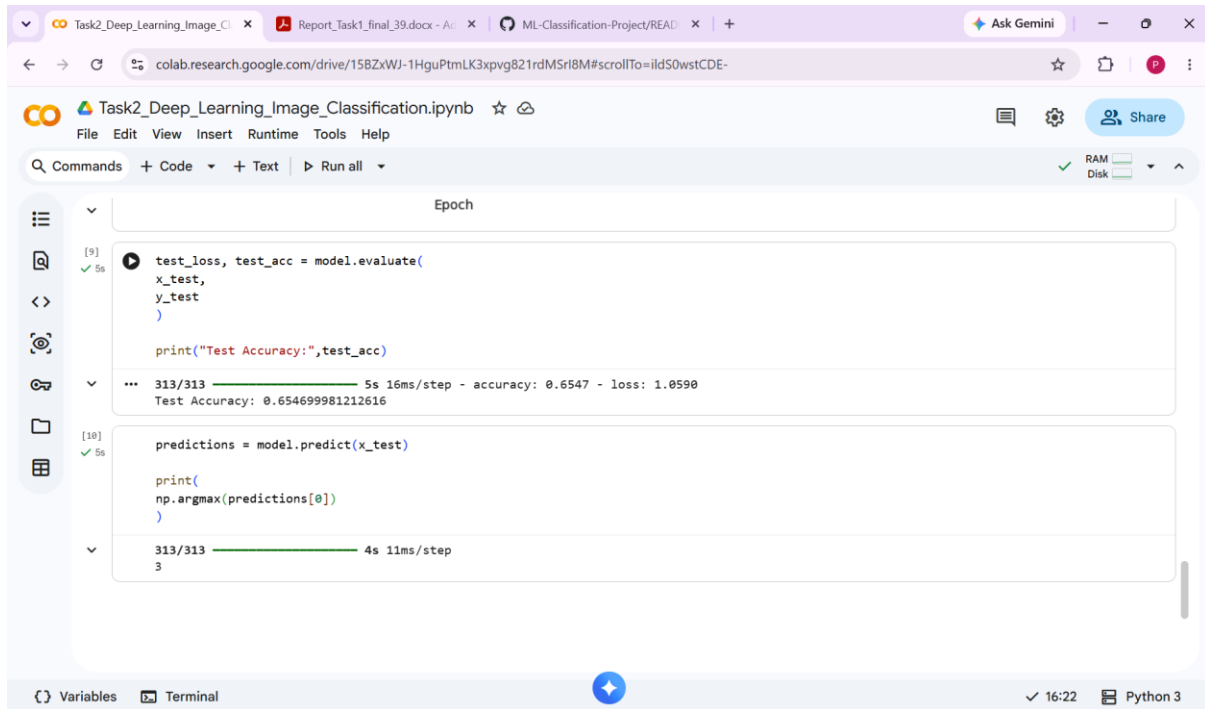
7. TRAINING CURVE

Training and validation accuracy curves were plotted to observe the model performance during training.



8. MODEL EVALUATION

The trained model was evaluated using testing data.



The screenshot shows a Google Colab notebook titled "Task2_Deep_Learning_Image_Classification.ipynb". The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with "Commands", "+ Code", "+ Text", and "Run all", and a status bar at the bottom showing "Variables", "Terminal", "16:22", and "Python 3".

The notebook contains two code cells. The first cell, labeled "[9]", contains the following code:

```
test_loss, test_acc = model.evaluate(
    x_test,
    y_test
)

print("Test Accuracy:", test_acc)
```

The output of this cell shows a progress bar for 313/313 steps, with a time of 5s and 16ms/step. The accuracy is 0.6547 and the loss is 1.0590. The final output is "Test Accuracy: 0.654699981212616".

The second cell, labeled "[10]", contains the following code:

```
predictions = model.predict(x_test)

print(
    np.argmax(predictions[0])
)
```

The output of this cell shows a progress bar for 313/313 steps, with a time of 4s and 11ms/step. The final output is the integer "3".

9. CONCLUSION

The CNN model successfully classified images into different categories using deep learning techniques. The model achieved good prediction accuracy and demonstrated the effectiveness of CNN architecture in image classification tasks.
